

CAP747:PROBABILITY AND STATISTICS

✓ Frequency distribution

Frequency distribution

A frequency distribution shows the **frequency of repeated items** in a graphical form or tabular form. It gives a visual display of the frequency of items or shows the number of times they occurred.

Frequency distribution is used to organize the collected data in table form. The data could be marks scored by students, temperatures of different towns, points scored in a volleyball match, etc. After data collection, we have to show data in a meaningful manner for better understanding. Organize the data in such a way that all its features are summarized in a table. This is known as frequency distribution.

Let's consider an example to understand this better. The following are the scores of 10 students in the G.K. quiz released by **Mr. Chris 15, 17, 20, 15, 20, 17, 17, 14, 14, 20**. Let's represent this data in frequency distribution and find out the number of students who got the same marks.

Quiz Marks	No. of Students
15	2
17	3
20	3
14	2

Frequency Distribution Graphs

- Bar graphs
- Histogram
- Pie chart
- Frequency Polygon

Types of Frequency Distribution

a) Grouped frequency distribution: In this type, the data is arranged and separated into groups called class intervals. The frequency of data belonging to each class interval is noted in a frequency distribution table. The grouped frequency table shows the distribution of frequencies in class intervals.

Suppose we conduct a survey in which we ask 15 households how many pets they have in their home. The results are as follows:

1, 1, 1, 1, 2, 2, 2, 3, 3, 4, 5, 5, 6, 7, 8

grouped frequency distribution for survey data:

Number of Pets	Frequency
1-2	7
3-4	3
5-6	3
7-8	2

We first created groups of size 2, then we counted how many **individual observations** from the dataset fell in each group. For example:

7 families had either 1 or 2 pets
3 families had either 3 or 4 pets
3 families had either 5 or 6 pets
2 families had either 7 or 8 pets

b)Ungrouped frequency distribution: It shows the frequency of an item in each separate data value rather than groups of data values.

1, 1, 1, 1, 2, 2, 2, 3, 3, 4, 5, 5, 6, 7, 8

ungrouped frequency distribution for our survey data:

Number of Pets	Frequency
1	4
2	3
3	2
4	1
5	2
6	1
7	1
8	1

This type of frequency distribution allows us to directly see how often different values occurred in our dataset. **individual observations** .For example:

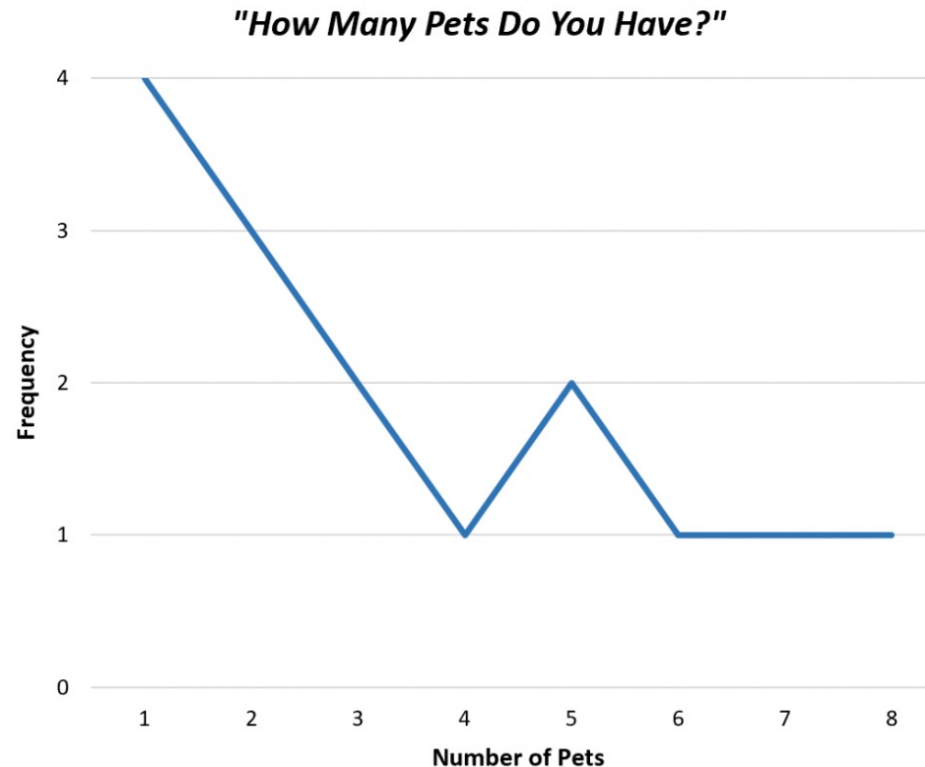
4 families had 1 pet
3 families had 2 pets
2 families had 3 pets
1 family had 4 pets
And so on.

Find

**When to Use Ungrouped Frequency
Distributions ???**

How to Visualize Ungrouped Frequency Distributions

The easiest way to visualize the values in an ungrouped frequency distribution is to create a **frequency polygon**, which displays the frequencies of each individual value in a simple chart.



Cumulative frequency

In Statistics, a cumulative frequency is defined as the total of frequencies, that are distributed over different class intervals. It means that the data and the total are represented in the form of a table in which the frequencies are distributed according to the class interval.

Types of Cumulative Frequency Distribution

- a) Less Than Cumulative Frequency**
- b) Greater Than Cumulative Frequency**

Example 1: Robert is the sales manager of a toy company. On checking his quarterly sales record, he can observe that by the month of April, a total of 83 toy cars were sold.

Month	Number of toy cars sold (Frequency)	Total number of toy cars sold (Cumulative Frequency)
January	20	20
February	30	$20 + 30 = 50$
March	15	$50 + 15 = 65$
April	18	$65 + 18 = 83$

Types of Cumulative Frequency Distribution

a) Less Than Cumulative Frequency

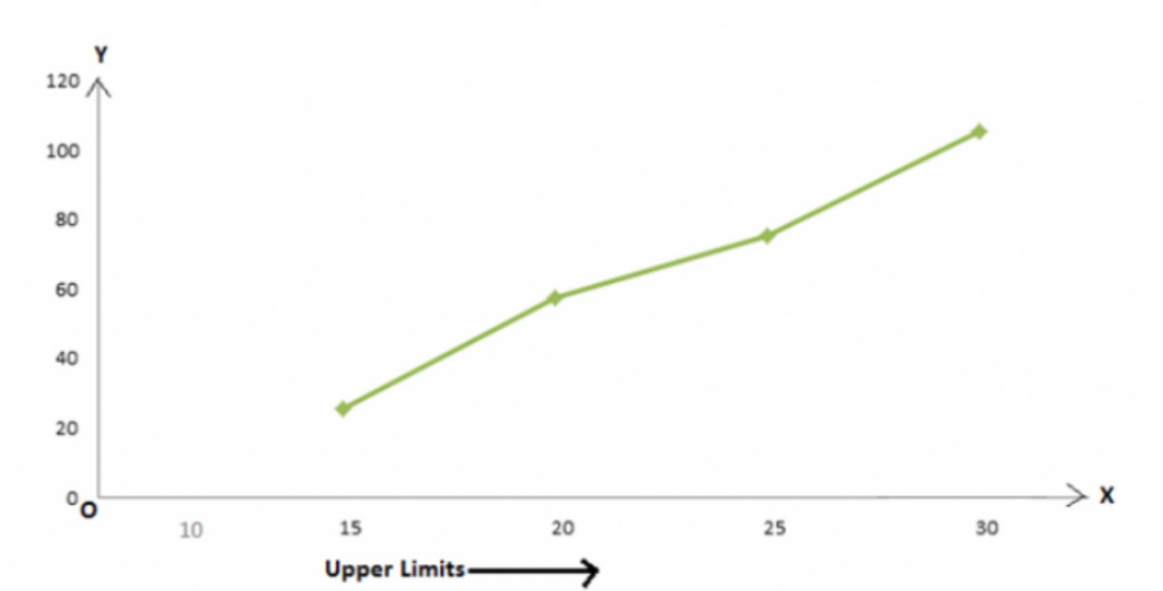
b) Greater Than Cumulative Frequency

Less than Cumulative

Age group	Number of participants (Frequency)
Less than 15	20
Less than 20	32
Less than 25	18
Less than 30	30



Age Group (class interval)	Cumulative Frequency
10-15	20
15-20	52
20-25	70
25-30	100

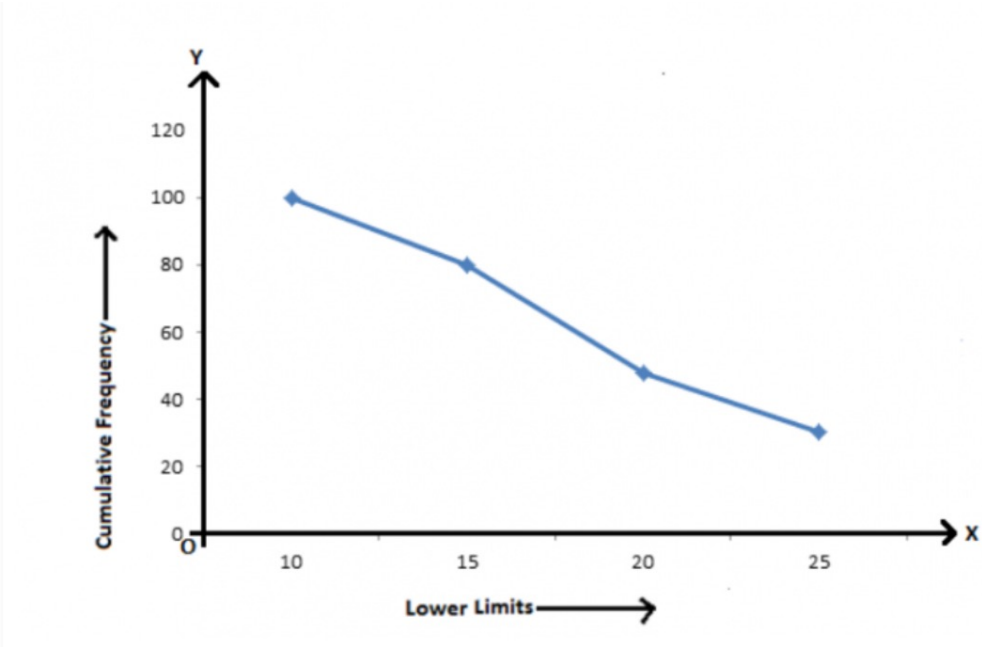


Greater Than Cumulative Frequency

Age group	Number of participants (Frequency)
More than 10	20
More than 15	32
More than 20	18
More than 25	30



Age Group (class interval)	Cumulative Frequency
10-30	100
15-30	80
20-30	48
25-30	30



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Solve :

Create a cumulative frequency table for the following information, which represent the number of hours per week that Arjun plays indoor games:

Arjun’s game time:

Days	No. of Hours
Monday	2 hrs
Tuesday	1 hr
Wednesday	2 hrs
Thursday	3 hrs
Friday	4 hrs
Saturday	2 hrs
Sunday	6 hrs